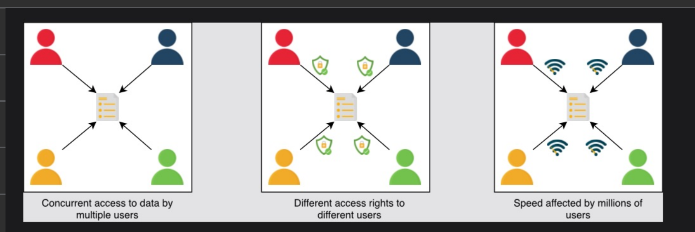
⑨ Intro to Databases

eg: Whatsapp : How do you store people & their messages?

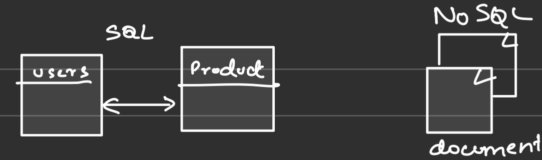
Limitations of file storage

- We can't offer concurrent management to separate users accessing the storage files from different locations.
- We can't grant different access rights to different users.
- How will the system scale and be available when adding thousands of entries?
- How will we search content for different users in a short time?



The limitations of file storage

- DB: organized collection of data, managed and accessed easily.
- store, retrive, modify & delete data
  eg: bank, online-shopping



- RDBMS : organized, predetermined schema
- NoSQL : file dir : unstructured, scatterd, dynamicsche

Advantages

A proper database is essential for every business or organization. This is because the database stores all essential information about the organization, such as personnel records, transactions, salary information, and so on. Following are some of the reasons why the database is important:

- Managing large data: A large amount of data can be easily handled with a database, which wouldn't be possible using other tools.
- Retrieving accurate data (data consistency): Due to different constraints in databases, we can retrieve accurate data whenever we want.
- Easy updation: It is quite easy to update data in databases using data manipulation language (DML).
- Security: Databases ensure the security of the data. A database only allows authorized users to access data.
- Data integrity: Databases ensure data integrity by using different constraints for data.
- Availability: Databases can be replicated (using data replication) on different servers, which can be concurrently updated. These replicas ensure availability.
- Scalability: Databases are divided (using data partitioning) to manage the load on a single node. This increases scalability.

★ **Types of DB**
- RDBMS
- Non-RDB

★ Relational DB :

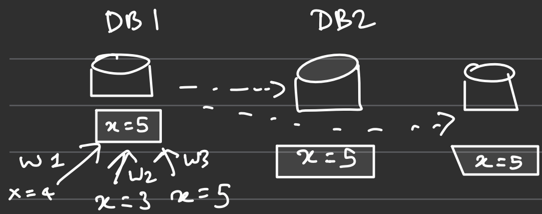
- rigid schema, prior structure
- data is organized into relation (table)
  unique key for each tuple (row/instance)
- each entity of table has attributes
  where instances are stored into row.
- FK : a tuple in one table is linked to a table in another table
- SQL : manipulating the DB . CURD.
- ACID : hides dirty read/write, read skew, lost updates, phantom reads) behind transactions

→ Atomicity : Each transaction is an atomic unit.
∴ all elements in trans execute or none.
If one stmt fails, all fails.


SUCCESS!


FAIL!

→ Consistency : all DB should be in consistent state after every transactions



→ Isolation : In case of multiple transactions, it shouldnot affect another.
The final state = same as the trans that were executed sequentially

eg :  Lock  unlock.

→ Durability : guarantee that completed trans will survive permanently

eg: $\boxed{\text{Commit}}$

eg : • MySQL • Oracle Database • SQL Server
• IBM DB2, • Postgres • SQLite

Why relational databases?

Relational databases are the default choices of software professionals for structured data storage. There are a number of advantages to these databases. One of the greatest powers of the relational database is its abstractions of ACID transactions and related programming semantics. This makes it very convenient for the end programmer to use a relational database. Let's revisit some important features of relational databases:

Flexibility

In the context of SQL, data definition language (DDL)provides us the flexibility to modify the database, including tables, columns, renaming the tables, and other changes. DDL even allows us to modify schema while other queries are happening and the database server is running.

Reduced redundancy

One of the biggest advantages of the relational database is that it eliminates data redundancy. The information related to a specific entity appears in one table while the relevant data to that specific entity appears in the other tables linked through foreign keys. This process is called normalization and has the additional benefit of removing an inconsistent dependency.

Concurrency

Concurrency is an important factor while designing an enterprise database. In such a case, the data is read and written by many users at the same time. We need to coordinate such interactions to avoid inconsistency in data—for example, the double booking of hotel rooms. Concurrency in a relational database is handled through transactional access to the data. As explained earlier, a transaction is considered an atomic operation, so it also works in error handling to either roll back or commit a transaction on successful execution.

Integration

The process of aggregating data from multiple sources is a common practice in enterprise applications. A common way to perform this aggregation is to integrate a shared database where multiple applications store their data. This way, all the applications can easily access each other's data while the concurrency control measures handle the access of multiple applications.

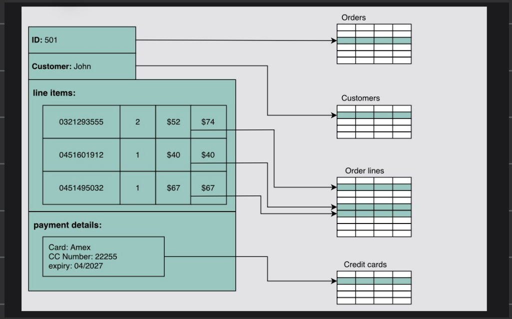
Backup and disaster recovery

Relational databases guarantee the state of data is consistent at any time. The export and import operations make backup and restoration easier. Most cloud-based relational databases perform continuous mirroring to avoid loss of data and make the restoration process easier and quicker.

Drawback

Impedance mismatch

Impedance mismatch is the difference between the relational model and the in-memory data structures. The relational model organizes data into a tabular structure with relations and tuples. SQL operation on this structured data yields relations aligned with relational algebra. However, it has some limitations. In particular, the values in a table take simple values that can't be a structure or a list. The case is different for in-memory, where a complex data structure can be stored. To make the complex structures compatible with the relations, we would need a translation of the data in light of relational algebra. So, the impedance mismatch requires translation between two representations, as denoted in the following figure:



NoSQL

- variety of data models to access & manage data
- semi-structured & unstructured data
- low latency, flexible data models
- character:

Simple design: Unlike relational databases, NoSQL doesn't require dealing with the impedance mismatch—for example, storing all the employees' data in one document instead of multiple tables that require join operations. This strategy makes it simple and easier to write less code, debug, and maintain.

Horizontal scaling: Primarily, NoSQL is preferred due to its ability to run databases on a large cluster. This solves the problem when the number of concurrent users increases. NoSQL makes it easier to scale out since the data related to a specific employee is stored in one document instead of multiple tables over nodes. NoSQL databases often spread data across multiple nodes and balance data and queries across nodes automatically. In case of a node failure, it can be transparently replaced without any application disruption.

Availability: To enhance the availability of data, node replacement can be performed without application downtime. Most of the non-relational databases' variants support data replication to ensure high availability and disaster recovery.

Support for unstructured and semi-structured data: Many NoSQL databases work with data that doesn't have schema at the time of database configuration or data writes. For example, document databases are structureless; they allow documents (JSON, XML, BSON, and so on) to have different fields. For example, one JSON document can have fewer fields than the other.

Cost: Licenses for many RDBMSs are pretty expensive, while many NoSQL databases are open source and freely available. Similarly, some RDBMSs rely on costly proprietary hardware and storage systems, while NoSQL databases usually use clusters of cheap commodity servers.

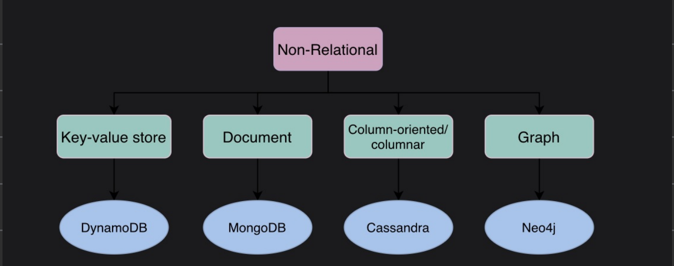
Types:

1) Key-value database

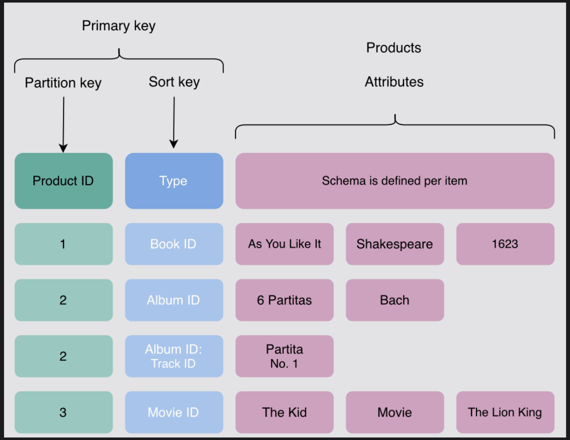
Key-value databases use key-value methods like hash tables to store data in key-value pairs. We can see this depicted in the figure a couple of paragraphs below. Here, the key serves as a unique or primary key, and the values can be anything ranging from simple scalar values to complex objects. These databases allow easy partitioning and horizontal scaling of the data. Some popular key-value databases include Amazon DynamoDB, Redis, and Memcached DB.

Use case: Key-value databases are efficient for session-oriented applications. Session oriented-applications, such as web applications, store users' data in the main memory or in a database during a session. This data may include user profile information, recommendations, targeted promotions, discounts, and more. A unique ID (a key) is assigned to each user's session for easy access and storage. Therefore, a better choice to store such data is the key-value database.

The following figure shows an example of a key-value database. The Product ID and Type of the item are collectively considered as the primary key. This is considered as a key for this key-value database. Moreover, the schema for storing the item attributes is defined based on the nature of the item and the number of attributes it possesses.



The types of NoSQL databases



Data stored in the form of key-value pair in DynamoDB, where the key is the combination of two attributes (Product ID and Type)

ii) Document database

A **document database** is designed to store and retrieve documents in formats like XML, JSON, BSON, and so on. These documents are composed of a hierarchical tree data structure that can include maps, collections, and scalar values. Documents in this type of database may have varying structures and data. MongoDB and Google Cloud Firestore are examples of document databases.

Use case: Document databases are suitable for unstructured catalog data, like JSON files or other complex structured hierarchical data. For example, in e-commerce applications, a product has thousands of attributes, which is unfeasible to store in a relational database due to its impact on the reading performance. Here comes the role of a document database, which can efficiently store each attribute in a single file for easy management and faster reading speed. Moreover, it's also a good option for content management applications, such as blogs and video platforms. An entity required for the application is stored as a single document in such applications.

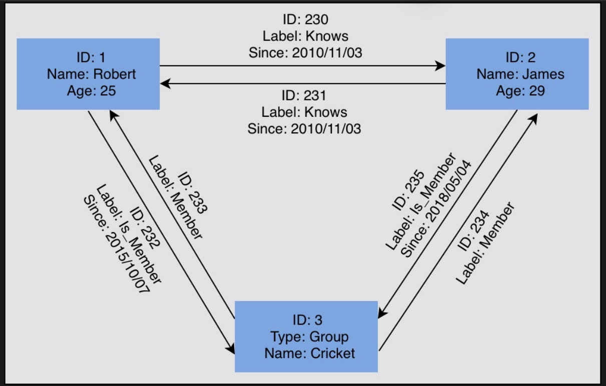
The following example shows data stored in a JSON document. This data is about a person. Various attributes are stored in the file, including `id`, `name`, `email`, and so on.

```
{ "id" : 1001,
  "name" : "Brown",
  "title" : "Mr."
  "likes" : [
        "design"
        "cycling"],
```

iii) Graph database

Graph databases use the graph data structure to store data, where nodes represent entities, and edges show relationships between entities. The organization of nodes based on relationships leads to interesting patterns between the nodes. This database allows us to store the data once and then interpret it differently based on relationships. Popular graph databases include Neo4J, OrientDB, and InfiniteGraph. Graph data is kept in store files for persistent storage. Each of the files contains data for a specific part of the graph, such as nodes, links, properties, and so on.

In the following figure, some data is stored using a graph data structure in nodes connected to each other via edges representing relationships between nodes. Each node has some properties, like `Name`, `ID`, and `Age`. The node having `ID: 2` has the `Name` of `James` and `Age` of `29` years.



A graph consists of nodes and links. This graph captures entities and their relationships with each other

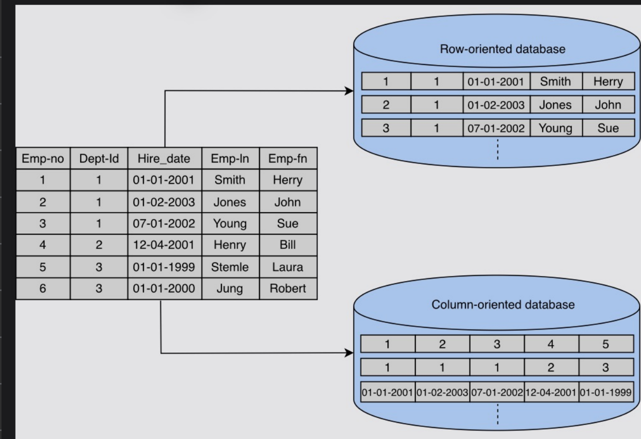
Use case: Graph databases can be used in social applications and provide interesting facts and figures among different kinds of users and their activities. The focus of graph databases is to store data and pave the way to drive analyses and decisions based on relationships between entities. The nature of graph databases makes them suitable for various applications, such as data regulation and privacy, machine learning research, financial services-based applications, and many more.

iv) Columnar database

Columnar databases store data in columns instead of rows. They enable access to all entries in the database column quickly and efficiently. Popular columnar databases include HBase, Hypertable, and Amazon Redshift.

Use case: Columnar databases are efficient for a large number of aggregation and data analytics queries. It drastically reduces the disk I/O requirements and the amount of data required to load from the disk. For example, in applications related to financial institutions, there's a need to sum the financial transaction over a period of time. Columnar databases make this operation quicker by just reading the column for the amount of money, ignoring other attributes of customers.

The following figure shows an example of a columnar database, where data is stored in a column-oriented format. This is unlike relational databases, which store data in a row-oriented fashion:



Column-oriented and row-oriented database

Drawback :

Lack of standardization

NoSQL doesn't follow any specific standard, like how relational databases follow relational algebra. Porting applications from one type of NoSQL database to another might be a challenge.

Consistency

NoSQL databases provide different products based on the specific trade-offs between consistency and availability when failures can happen. We won't have strong data integrity, like primary and referential integrities in a relational database. Data might not be strongly consistent but slowly converging using a weak model like eventual consistency.

- key-value eg: persist user state, memcached/s Redis used for caching in use cases
- doc db - web based multiplayer game. Real time feed
- NOSQL → FB
- tabular → relational

DATA REPLECATION

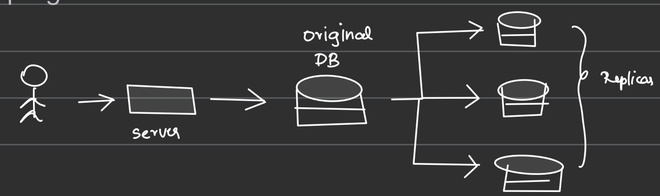
Data is an asset for an organization because it drives the whole business. Data provides critical business insights into what's important and what needs to be changed.
Organizations also need to securely save and serve their clients' data on demand. Timely access to the required data under varying conditions (increasing reads and writes, disks and node failures, network and power outages, and so on) is required to successfully run an online business.
We need the following characteristics from our data store:

- Availability under faults (failure of some disk, nodes, and network and power outages).
- Scalability (with increasing reads, writes, and other operations).
- Performance (low latency and high throughput for the clients).

It's challenging, or even impossible, to achieve the above characteristics on a single node.

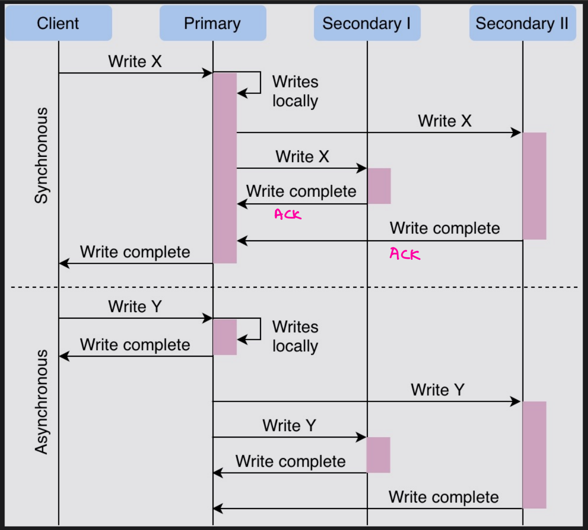
REPLECATION:

Replication refers to keeping multiple copies of the data at various nodes (preferably geographically distributed) to achieve availability, scalability, and performance. In this lesson, we assume that a single node is enough to hold our entire data. We won't use this assumption while discussing the partitioning of data in multiple nodes. Often, the concepts of replication and partitioning go together.
However, with many benefits, like availability, replication comes with its complexities. Replication is relatively simple if the replicated data doesn't require frequent changes. The main problem in replication arises when we have to maintain changes in the replicated data over time.
Additional complexities that could arise due to replication are as follows:
How do we keep multiple copies of data consistent with each other?
How do we deal with failed replica nodes?
Should we replicate synchronously or asynchronously?
How do we deal with replication lag in case of asynchronous replication?
How do we handle concurrent writes?
What consistency model needs to be exposed to the end programmers?



* SYNCH vs ASYNCH REPLICATION :

In **synchronous replication**, the primary node waits for acknowledgments from secondary nodes about updating the data. After receiving acknowledgment from all secondary nodes, the primary node reports success to the client. Whereas in **asynchronous replication**, the primary node doesn't wait for the acknowledgment from the secondary nodes and reports success to the client after updating itself.
The advantage of synchronous replication is that all the secondary nodes are completely up to date with the primary node. However, there's a disadvantage to this approach. If one of the secondary nodes doesn't acknowledge due to failure or fault in the network, the primary node would be unable to acknowledge the client until it receives the successful acknowledgment from the crashed node. This causes high latency in the response from the primary node to the client.
On the other hand, the advantage of asynchronous replication is that the primary node can continue its work even if all the secondary nodes are down. However, if the primary node fails, the writes that weren't copied to the secondary nodes will be lost.
The above paragraph explains a trade-off between data consistency and availability when different components of the system can fail.



Synchronous versus asynchronous replication

Data replication models

Now, let's discuss various mechanisms of data replication. In this section, we'll discuss the following models along with their strengths and weaknesses:

1. Single leader or primary-secondary replication
2. Multi-leader replication
3. Peer-to-peer or leaderless replication

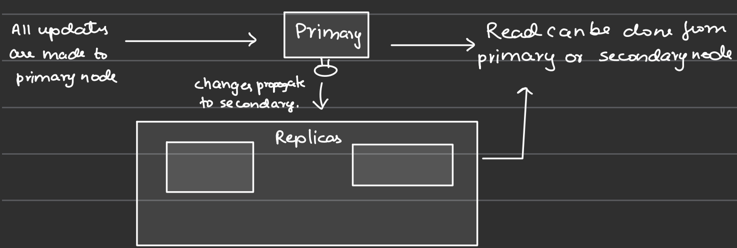
Single Leader Replication:

In **primary-secondary replication,** data is replicated across multiple nodes. One node is designated as the primary. It's responsible for processing any writes to data stored on the cluster. It also sends all the writes to the secondary nodes and keeps them in sync.

Primary-secondary replication is appropriate when our workload is read-heavy. To better scale with increasing readers, we can add more followers and distribute the read load across the available followers. However, replicating data to many followers can make a primary bottleneck. Additionally, primary-secondary replication is inappropriate if our workload is write-heavy.

Another advantage of primary-secondary replication is that it's read resilient. Secondary nodes can still handle read requests in case of primary node failure. Therefore, it's a helpful approach for read-intensive applications.

Replication via this approach comes with inconsistency if we use asynchronous replication. Clients reading from different replicas may see inconsistent data in the case of failure of the primary node that couldn't propagate updated data to the secondary nodes. So, if the primary node fails, any missed updates not passed on to the secondary nodes can be lost.



Let's discuss each of them in detail.

Statement-based replication

Statement-based replication (SBR) is an approach used in MySQL databases. In this approach, the primary node executes the SQL statements such as INSERT, UPDATE, DELETE, etc., and then the statements are written into a log file. In the next step, the log file is sent to the secondary nodes for execution. This type of replication was used in MySQL before version 5.1.

While this type of replication seems good, it also has some disadvantages. For example, any nondeterministic functions such as NOW() might result in distinct writes on the primary and secondary nodes.

Write-ahead log (WAL) shipping

Write-ahead log (WAL) shipping is a data replication technique used in both PostgreSQL and Oracle. In this technique, when a transaction occurs, it's initially recorded in a transactional log file, and the log file is written to disk. Subsequently, the recorded operations are executed on the primary database before being transmitted to secondary nodes for execution. Unlike SBR, WAL maintains transactional logs instead of SQL statements into a log file, ensuring consistency when dealing with nondeterministic functions. Writing to disk also aids in recovery in case of crash failures.

For example, when an operation like an `UPDATE` is executed in PostgreSQL, it's first written to the transactional log file and disk before being applied to the database. This entry in the transactional log can include details such as the transaction ID, operation type, affected table, and new values, after which the changes are replicated to the secondary nodes. However, the drawback of WAL is its tight coupling with the inner structure of the database engine, making software upgrades on the leader and followers complicated.

Logical (row-based) replication

Logical (row-based) replication is utilized in various relational databases, including PostgreSQL and MySQL. In this approach, changes made to the database are captured at the level of individual rows and then replicated to the secondary nodes. Instead of replicating the actual physical changes made to the database, this approach captures the operations in a logical format and then executes them on secondary nodes.

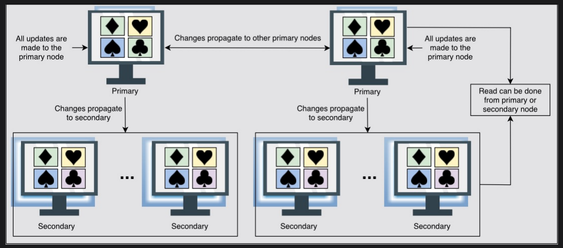
For example, when operations like `INSERT` or `UPDATE` are performed, the entire affected row is captured on the primary node, containing all the column values of the specified row. This captured change is then executed on secondary nodes, where they ensure that the data remains consistent with the data on the primary node. It offers advantages in terms of flexibility and compatibility with different types of schemas.

Multi-leader replication

As discussed above, single leader replication using asynchronous replication has a drawback. There's only one primary node, and all the writes have to go through it, which limits the performance. In case of failure of the primary node, the secondary nodes may not have the updated database.

Multi-leader replication is an alternative to single leader replication. There are multiple primary nodes that process the writes and send them to all other primary and secondary nodes to replicate. This type of replication is used in databases along with external tools like the Tungsten Replicator for MySQL.

This kind of replication is quite useful in applications in which we can continue work even if we're offline—for example, a calendar application in which we can set our meetings even if we don't have access to the internet. Once we're online, it replicates its changes from our local database (our mobile phone or laptop acts as a primary node) to other nodes



Multi-leader data replication model (all-to-all topology)

Conflict

Multi-leader replication gives better performance and scalability than single leader replication, but it also has a significant disadvantage. Since all the primary nodes concurrently deal with the write requests, they may modify the same data, which can create a conflict between them. For example, suppose the same data is edited by two clients simultaneously. In that case, their writes will be successful in their associated primary nodes, but when they reach the other primary nodes asynchronously, it creates a conflict.

Handle conflicts

Conflicts can result in different data at different nodes. These should be handled efficiently without losing any data. Let's discuss some of the approaches to handle conflicts:

Conflict avoidance

A simple strategy to deal with conflicts is to prevent them from happening in the first place. Conflicts can be avoided if the application can verify that all writes for a given record go via the same leader.

However, the conflict may still occur if a user moves to a different location and is now near a different data center. If that happens, we need to reroute the traffic. In such scenarios, the conflict avoidance approach fails and results in concurrent writes.

Last-write-wins

Using their local clock, all nodes assign a timestamp to each update. When a conflict occurs, the update with the latest timestamp is selected.

This approach can also create difficulty because the clock synchronization across nodes is challenging in distributed systems. There's clock skew that can result in data loss.

Custom logic

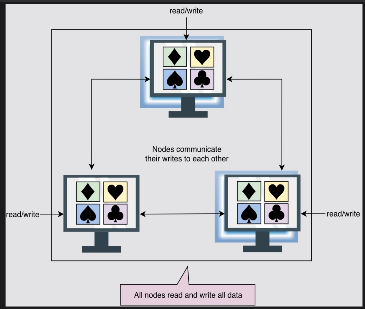
In this approach, we can write our own logic to handle conflicts according to the needs of our application. This custom logic can be executed on both reads and writes. When the system detects a conflict, it calls our custom conflict handler.

Multi-leader replication topologies

There are many topologies through which multi-leader replication is implemented, such as circular topology, star topology, and all-to-all topology. The most common is the all-to-all topology. In star and circular topology, there's again a similar drawback that if one of the nodes fails, it can affect the whole system. That's why all-to-all is the most used topology.

Peer-to-peer/leaderless replication

In primary-secondary replication, the primary node is a bottleneck and a single point of failure. Moreover, it helps to achieve read scalability but fails to provide write scalability. The **peer-to-peer replication** model resolves these problems by not having a single primary node. All the nodes have equal weightage and can accept read and write requests. This replication scheme can be found in the Cassandra database.



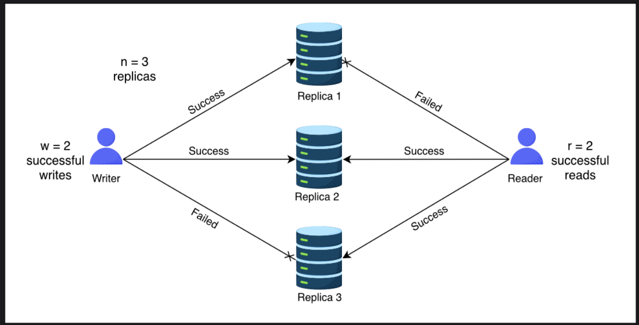
Peer-to-peer data replication model where all nodes apply reads and writes to all the data

Like primary-secondary replication, this replication can also yield inconsistency. This is because when several nodes accept write requests, it may lead to concurrent writes. A helpful approach used for solving write-write inconsistency is called **quorums**.

Quorums

Let's suppose we have three nodes. If at least two out of three nodes are guaranteed to return successful updates, it means only one node has failed. This means that if we read from two nodes, at least one of them will have the updated version, and our system can continue working.

If we have n nodes, then every write must be updated in at least w nodes to be considered a success, and we must read from r nodes. We'll get an updated value from reading as long as $w + r > n$ because at least one of the nodes must have an updated write from which we can read. Quorum reads and writes adhere to these r and w values. These n, w, and r are configurable in Dynamo-style databases.



Reader getting an updated value from replica 2

- Read heavy – high throughput, low latency to client, low implementation heavy

DATA PARTITIONIG

Data is an asset for any organization. Increasing data and concurrent read/write traffic to the data puts scalability pressure on traditional databases. As a result, the latency and throughput are affected. Traditional databases are attractive due to their properties such as range queries, secondary indices, and transactions with the ACID properties.

At some point, a single node-based database isn't enough to tackle the load. We might need to distribute the data over many nodes but still export all the nice properties of relational databases. In practice, it has proved challenging to provide single-node database-like properties over a distributed database.

One solution is to move data to a NoSQL-like system. However, the historical codebase and its close cohesion with traditional databases make it an expensive problem to tackle. Organizations might scale traditional databases by using a third-party solution. But often, integrating a third-party solution has its complexities. More importantly, there are abundant opportunities to optimize for the specific problem at hand and get much better performance than a general-purpose solution. Data partitioning (or sharding) enables us to use multiple nodes where each node manages some part of the whole data. To handle increasing query rates and data amounts, we strive for balanced partitions and balanced read/write load.

Sharding

To divide load among multiple nodes, we need to partition the data by a phenomenon known as **partitioning** or **sharding**. In this approach, we split a large dataset into smaller chunks of data stored at different nodes on our network.

The partitioning must be balanced so that each partition receives about the same amount of data. If partitioning is unbalanced, the majority of queries will fall into a few partitions. Partitions that are heavily loaded will create a system bottleneck. The efficacy of partitioning will be harmed because a significant portion of data retrieval queries will be sent to the nodes that carry the highly congested partitions. Such partitions are known as hotspots. Generally, we use the following two ways to shard the data:

Vertical sharding

Horizontal sharding

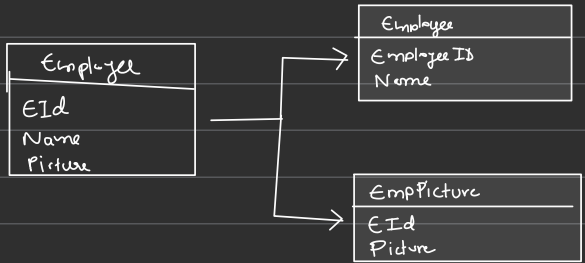
Vertical sharding

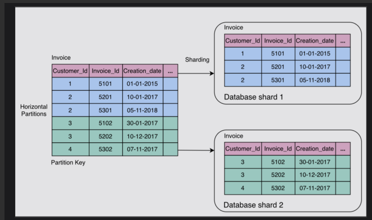
We can put different tables in various database instances, which might be running on a different physical server. We might break a table into multiple tables so that some columns are in one table while the rest are in the other. We should be careful if there are joins between multiple tables. We may like to keep such tables together on one shard.

Often, **vertical sharding** is used to increase the speed of data retrieval from a table consisting of columns with very wide text or a binary large object (blob). In this case, the column with large text or a blob is split into a different table.

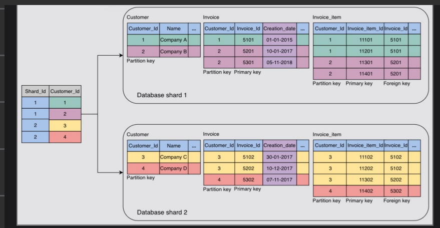
As shown in the figure a couple paragraphs below, the `Employee`table is divided into two tables: a reduced `Employee` table and an `EmployeePicture` table. The `EmployeePicture` table has just two columns, `EmployeeID` and `Picture`, separated from the original table. Moreover, the primary key `EmployeeID` of the `Employee` table is added in both partitioned tables. This makes the data read and write easier, and the reconstruction of the table is performed efficiently.

Vertical sharding has its intricacies and is more amenable to manual partitioning, where stakeholders carefully decide how to partition data. In comparison, horizontal sharding is suitable to automate even under dynamic conditions.





Horizontal partitioning



Horizontal partitioning on a set of tables

Horizontal sharding

At times, some tables in the databases become too big and affect read/write latency. **Horizontal sharding** or partitioning is used to divide a table into multiple tables by splitting data row-wise, as shown in the figure in the next section. Each partition of the original table distributed over database servers is called a **shard**. Usually, there are two strategies available:

Key-range based sharding

Hash based sharding

Key-range based sharding

In the **key-range based sharding**, each partition is assigned a continuous range of keys.

In the following figure, horizontal partitioning on the `Invoice` table is performed using the key-range based sharding with `Customer_Id` as the partition key. The two different colored tables represent the partitions.

Sometimes, a database consists of multiple tables bound by foreign key relationships. In such a case, the horizontal partition is performed using the same partition key on all tables in a relation. Tables (or subtables) that belong to the same partition key are distributed to one database shard. The following figure shows that several tables with the same partition key are placed in a single database shard:

The basic design techniques used in multi-table sharding are as follows:

There's a partition key in the `Customer` mapping table. This table resides on each shard and stores the partition keys used in the shard. Applications create a mapping logic between the partition keys and database shards by reading this table from all shards to make the mapping efficient. Sometimes, applications use advanced algorithms to determine the location of a partition key belonging to a specific shard.

The partition key column, Customer_Id, is replicated in all other tables as a data isolation point. It has a trade-off between an impact on increased storage and locating the desired shards efficiently. Apart from this, it’s helpful for data and workload distribution to different database shards. The data routing logic uses the partition key at the application tier to map queries specified for a database shard.
Primary keys are unique across all database shards to avoid key collision during data migration among shards and the merging of data in the online analytical processing (OLAP)environment.
The column Creation_date serves as the data consistency point, with an assumption that the clocks of all nodes are synchronized. This column is used as a criterion for merging data from all database shards into the global view when essential.

Advantages

- Using key-range-based sharding method, the range-query-based scheme is easy to implement. We precisely know where (which node, which shard) to look for a specific range of keys.
- Range queries can be performed using the partitioning keys, and those can be kept in partitions in sorted order. How exactly such a sorting happens over time as new data comes in is implementation specific.

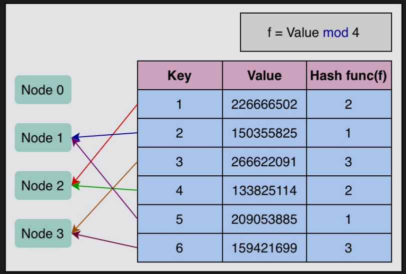
Disadvantages

- Range queries can’t be performed using keys other than the partitioning key.
- If keys aren’t selected properly, some nodes may have to store more data due to an uneven distribution of the traffic.

Hash-based sharding

Hash-based sharding uses a hash function on an attribute. This hash function produces a hash value that is used to perform partitioning. The main concept is to use a hash function on the key to get a hash value and then mod by the number of partitions. Once we’ve found an appropriate hash function for keys, we may give each partition a range of hashes (rather than a range of keys). Any key whose hash occurs inside that range will be kept in that partition.

In the illustration below, we use a hash function of ValueValueValue modmodmod nnn. The nnn is the number of nodes, which is four. We allocate keys to nodes by checking the mod for each key. Keys with a mod value of 2 are allocated to node 2. Keys with a mod value of 1 are allocated to node 1. Keys with a mod value of 3 are allocated to node 3. Because there’s no key with a mod value of 0, node 0 is left vacant.



Hash-based sharding

Advantages

- Keys are uniformly distributed across the nodes.

Disadvantages

- We can’t perform range queries with this technique. Keys will be spread over all partitions.

Note: How many shards per database?

Empirically, we can determine how much each node can serve with acceptable performance. It can help us find out the maximum amount of data that we would like to keep on any one node. For example, if we find out that we can put a maximum of 50 GB of data on one node, we have the following:

Database size is 10 TB

Size of a single shard is 50 GB

Number of shards the database should be distributed in = 10 TB/50 GB = 200 shards

- Hashing maps data to a fixed-size value (a hash).
- Useful for fast data access and security.
- A good hash function avoids collisions (two inputs getting the same

Example:

Name: "Amy"

- A → ASCII 65
- m → ASCII 109
- y → ASCII 121

Sum = 65 + 109 + 121 = 295

Hash = 295 % 10 = 5

So, we store "Amy" at index 5 in an array or hash table.

Consistent hashing

Consistent hashing assigns each server or item in a distributed hash table a place on an abstract circle, called a ring, irrespective of the number of servers in the table. This permits servers and objects to scale without compromising the system's overall performance.

Advantages of consistent hashing

It's easy to scale horizontally.

It increases the throughput and improves the latency of the application.

Disadvantages of consistent hashing

Randomly assigning nodes in the ring may cause non-uniform distribution.

Rebalance the partitions

Query load can be imbalanced across the nodes due to many reasons, including the following:

The distribution of the data isn't equal.

There's too much load on a single partition.

There's an increase in the query traffic, and we need to add more nodes to keep up.

We can apply the following strategies to rebalance partitions.

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We can apply the following strategies to rebalance partitions.

Avoid hash mod n

Usually, we avoid the hash of a key for partitioning (we used such a scheme to explain the concept of hashing in simple terms earlier). The problem with the addition or removal of nodes in the case of hashmodnhash mod nhashmodn is that every node's partition number changes and a lot of data moves. For example, assume we have hash(key)=1235hash(key) = 1235hash(key)=1235. If we have five nodes at the start, the key will start on node 1 (123512351235modmodmod 5=05 = 05=0). Now, if a new node is added, the key would have to be moved to node 6 (123512351235 modmodmod 6=56 = 56=5), and so on. This moving of keys from one node to another makes rebalancing costly

Sure! Let's break it down super simply:

🔄 What is hash(key) % n?

Imagine you want to split your data across n nodes (computers). You take a key (like a user ID), convert it to a number using a hash function, and then use % n to decide which node gets that data.

Example:

hash("user123") = 1235
n = 5 nodes
1235 % 5 = 0 → Store on Node 0

❌ What's the Problem?

When you add or remove a node, the value of n changes — and almost every key moves to a different node.

Example:

- Before: 5 nodes → hash("user123") % 5 = 0
- Add a node (n = 6) → 1235 % 6 = 5

🔁 Now "user123" must move from Node 0 → Node 5.
And this happens to almost all your keys — that's very expensive when you have millions of them.

So Why Avoid hash % n?

Because it causes massive reshuffling of data whenever the number of nodes changes.

💡 What to Use Instead?

Use consistent hashing.

- Only a small portion of the keys move when a node is added/removed.
- Data movement is minimal, making rebalancing much cheaper and faster.

🧠 TL;DR

Approach	Problem When Adding/Removing Node
hash(key) % n	🔁 All keys get reshuffled (bad!)
Consistent Hashing	✅ Only a few keys move (good!)

Fixed number of partitions

In this approach, the number of partitions to be created is fixed at the time when we set our database up. We create a higher number of partitions than the nodes and assign these partitions to nodes. So, when a new node is added to the system, it can take a few partitions from the existing nodes until the partitions are equally divided.

There's a downside to this approach. The size of each partition grows with the total amount of data in the cluster since all the partitions contain a small part of the total data. If a partition is very small, it will result in too much overhead because we may have to make a large number of small-sized partitions, each costing us some overhead. If the partition is very large, rebalancing the nodes and recovering from node failures will be expensive. It's very important to choose the right number of partitions. A fixed number of partitions is used in Elasticsearch, Riak, and many more.

Dynamic partitioning

In this approach, when the size of a partition reaches the threshold, it's split equally into two partitions. One of the two split partitions is assigned to one node and the other one to another node. In this way, the load is divided equally. The number of partitions adapts to the overall data amount, which is an advantage of dynamic partitioning.

However, there's a downside to this approach. It's difficult to apply dynamic rebalancing while serving the reads and writes. Dynamic rebalancing during reads and writes is challenging because it involves moving data between nodes, causing latency and potential conflicts. Ensuring data consistency (as data is simultaneously moved and accessed) and availability (potentially requiring pauses in reads/writes during rebalancing) introduces complexities that can impact system performance and reliability. This approach is used in HBase and MongoDB

There are two ways to perform rebalancing: automatic and manual. In **automatic rebalancing**, there's no administrator. The system determines when to perform the partitions and when to move data from one node to another.

In **manual rebalancing**, the administrator determines when and how to perform the partitioning. Organizations perform rebalancing according to their needs. Some use automatic rebalancing, and some use manual.

Partition proportionally to nodes

In this approach, the number of partitions is proportionate to the number of nodes, which means every node has fixed partitions. In earlier approaches, the number of partitions was dependent on the size of the dataset. That isn't the case here. While the number of nodes remains constant, the size of each partition rises according to the dataset size. However, as the number of nodes increases, the partitions shrink. When a new node enters the network, it splits a certain number of current partitions at random, then takes one half of the split and leaves the other half alone. This can result in an unfair split. This approach is used by Cassandra and Ketama.

Partitioning and secondary indexes

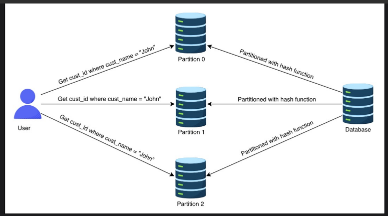
We've discussed key-value data model partitioning schemes in which the records are retrieved with primary keys. But what if we have to access the records through secondary indexes? Secondary indexes are the records that aren't identified by primary keys but are just a way of searching for some value. For example, the above illustration of horizontal partitioning contains the customer table, searching for all customers with the same creation year.

We can partition with secondary indexes in the following ways.

Partition secondary indexes by document

Each partition is fully independent in this indexing approach. Each partition has its secondary indexes covering just the documents in that partition. It's unconcerned with the data held in other partitions. If we want to write anything to our database, we need to handle that partition only containing the document ID we're writin

g. It's also known as the local index. In the illustration below, there are three partitions, each having its own identity and data. If we want to get all the customer IDs with the name John, we have to request from all partitions.
However, this type of querying on secondary indexes can be expensive. As a result of being restricted by the latency of a poor-performing partition, read query latencies may increase.



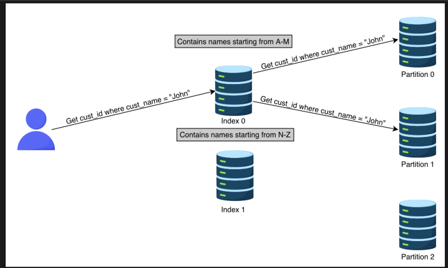
Partitioning secondary indexes by document

Partition secondary indexes by the term

Instead of creating a secondary index for each partition (a local index), we can make a global index for secondary terms that encompasses data from all partitions.

n the illustration below, we create indexes on names (the term on which we're partitioning) and store all the indexes for names on separated nodes. To get the `cust_id` of all the customers named `John`, we must determine where our term index is located. The `index 0` contains all the customers with names starting with "A" to "M." The `index 1` includes all the customers with names beginning with "N" to "Z." Because `John` lies in `index 0`, we fetch a list of `cust_id` with the name `John` from `index 0`.

Partitioning secondary indexes by the term is more read-efficient than partitioning secondary indexes by the document. This is because it only accesses the partition that contains the term. However, a single write in this approach affects multiple partitions, making the method write-intensive and complex.



Request routing

We've learned how to partition our data. However, one question arises here: How does a client know which node to connect to while making a request? The allocation of partitions to nodes varies after rebalancing. If we want to read a specific key, how do we know which IP address we need to connect to read?

This problem is also known as **service discovery**. Following are a few approaches to this problem:

- Allow the clients to request any node in the network. If that node doesn't contain the requested data, it forwards that request to the node that does contain the related data.
- The second approach contains a routing tier. All the requests are first forwarded to the routing tier, and it determines which node to connect to fulfill the request.
- The clients already have the information related to partitioning and which partition is connected to which node. So, they can directly contact the node that contains the data they need.

In all of these approaches, the main challenge is to determine how these components know about updates in the partitioning of the nodes.

ZooKeeper

To track changes in the cluster, many distributed data systems need a separate management server like ZooKeeper. **Zookeeper** keeps track of all the mappings in the network, and each node connects to ZooKeeper for the information. Whenever there's a change in the partitioning, or a node is added or removed, ZooKeeper gets updated and notifies the routing tier about the change. HBase, Kafka and SolrCloud use ZooKeeper.

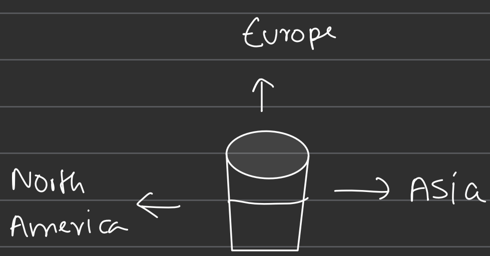
Let’s assess our understanding of what’s described in this lesson with the following question:

Imagine you’re a database architect for a rapidly expanding e-commerce platform with a global user base. The platform experiences varying user activity levels across different regions, as illustrated below. The existing monolithic database struggles to handle the increasing load. Users in different regions have distinct sets of preferences and tend to interact more within their regions. Now, we’re looking for an efficient and scalable solution to optimize performance, enhance scalability, and cater to regional variations in user behavior. Regarding this, which one of the following strategies would you adopt, and why?

Database sharding

Database replication

Note: Provide your answer in the following interactive widget:



The best strategy in this scenario is **database sharding**. Sharding involves horizontally partitioning the database into smaller, more manageable pieces called shards, each responsible for a subset of data. This approach allows the system to handle increased load efficiently, improves performance, and caters to regional variations by placing data closer to the users in each region.

While database replication can help with read scalability and fault tolerance, it doesn’t address the core issue of handling large volumes of data and high write loads across regions as effectively as sharding does.

Relational and Non-relational Databases

Relational Database	Non-relational Database
If the data to be stored is structured	If the data to be stored is unstructured
If ACID properties are required	If there’s a need to serialize and deserialize data
If the size of the data is relatively small and can fit on a node)	If the size of the data to be stored is large

Trade offs

Which is the best database sharding approach?

Both horizontal and vertical sharding involve adding resources to our computing infrastructure. Our business stakeholders must decide which is suitable for our organization. We must scale our resources accordingly for our organization and business to grow, to prevent downtime, and to reduce latency. We can scale these resources through a combination of adjustments to CPU, physical memory requirements, hard disk adjustments, and network bandwidth.

The following sections explain the pros and cons of centralized and distributed database:

Advantages and disadvantages of a centralized database

Advantages

- Data maintenance, such as updating and taking backups of a centralized database, is easy.
- Centralized databases provide stronger consistency and ACID transactions than distributed databases.
- Centralized databases provide a much simpler programming model for the end programmers as compared to distributed databases.
- It's more efficient for businesses that have a small amount of data to store that can reside on a single node.

Disadvantages

- A centralized database can slow down, causing high latency for end users, when the number of queries per second accessing the centralized database is approaching single-node limits.
- A centralized database has a single point of failure. Because of this, its probability of not being accessible is much higher

Advantages and disadvantages of a distributed database

Advantages

- It's fast and easy to access data in a distributed database because data is retrieved from the nearest database shard or the one frequently used.
- Data with different **levels of distribution transparency** can be stored in separate places.
- Intensive transactions consisting of queries can be divided into multiple optimized subqueries, which can be processed in a parallel fashion.

Disadvantages

- Sometimes, data is required from multiple sites, which takes more time than expected.
- Relations are partitioned vertically or horizontally among different nodes. Therefore, operations such as joins need to reconstruct complete relations by carefully fetching data. These operations can become much more expensive and complex.
- It's difficult to maintain consistency of data across sites in the distributed database, and it requires extra measures.
- Updations and backups in distributed databases take time to synchronize data.

Query optimization and processing speed in a distributed database

A transaction in the distributed database depends on the type of query, number of sites (shards) involved, communication speed, and other factors, such as underlying hardware and the type of database used. However, as an example, let's assume a query accessing three tables, `Store`, `Product`, and `Sales`, residing on different sites.